

Technical Datasheet TnH Sensor Modules with AOV/AOC

The **WIN-SN-TNH-AOC-AOV (Outdoor Edition)** is a robust sensor housed in a weather-resistant louvered enclosure, designed to accurately measure **Temperature and Humidity** in outdoor environments.

TnH Sensor Module: -



The device outputs precise **analog signals** with selectable **10-bit or 12-bit resolution**, supporting both **0-10V** and **4-20mA** ranges for easy integration with industrial systems. Powered by a standard **12-24V DC supply**, it features onboard **resettable fuses** to protect against reverse polarity on both the power and analog signal terminals. This resilient design makes it ideal for greenhouse climate control and general environmental monitoring.

Specifications

General –

Dimensions	158mm L x 139 mm D
Power	Input Power – 12 - 24 V DC
Operating Temperature	0 – 60° C (32 ~ 140°F)
Storage Temperature	-20 - 70° C (-4 ~ 158°F)
Storage Humidity	5 ~ 95 % RH, non – Condensing

Certifications



Technical Specifications: -

Measured Parameters	Temperature and Humidity
Output Type	Analog Output 10-bit / 12-bit resolution
Output Range	AOV 0-10V / AOC 4-20mA

	Sensor Name	Temperature Measuring Range	Humidity Measuring range	Operating Temperature	Measuring Accuracy
☐	AHT20	-20°C ~ 85°C.	0~100% RH.	-40°C ~ 120°C.	0.3°C, 2.0 %RH
☐	AHT21	-40°C ~ 120°C.	0~100% RH.	-40°C ~ 120°C.	0.3°C, 2.0 %RH

Applications

- Outdoor air quality monitoring.
- Climate control in greenhouses.
- Environmental monitoring systems.
- Smart home system.

Precautions

- Read all instructions before using the sensor board.
- Keep the sensor board away from water or moisture to prevent damage.
- Do not disassemble or modify the sensor board.
- Use only the specified power sources and mentioned.

Wire diagram:

ID	AO	Wire Colour	Parameter
1	AO1	Brown	Temperature +
2		Black	Temperature -
3	AO2	Yellow	Humidity +
4		White	Humidity -

Wire diagram:

Analog Output	Parameter	Range
AO1	Temperature	-40°C ~ 85°0043
AO2	Humidity	0 ~ 130 g/m ³

Contact us: -

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